Matrices, Vectors & Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the standard (x, y) coordinate plane, the complex number $z = x + yi$ satisfies the equation $z(2 - i) = 15 + 5i$ where $i^2 = -1$, and x and y are real numbers. Vector $\mathbf{v}$ is defined by $\mathbf{v} = [x; y]$. If matrix $A = [3, -1; 2, 4]$, what is the magnitude of the vector $A\mathbf{v}$? Show your complete algebraic work.

Final Answer: $10\sqrt{10}$. Step 1: Solve for the complex number $z = x + yi$. Multiply the numerator and denominator by the complex conjugate of $2 - i$, which is $2 + i$: $z = \frac{15 + 5i}{2 - i} = \frac{(15 + 5i)(2 + i)}{(2 - i)(2 + i)} = \frac{30 + 15i + 10i + 5i^2}{4 - i^2}$ Since $i^2 = -1$: $z = \frac{30 + 25i - 5}{4 - (-1)} = \frac{25 + 25i}{5} = 5 + 5i$ Thus, $x = 5$ and $y = 5$. Step 2: Construct vector $\mathbf{v}$ and perform the matrix-vector multiplication $A\mathbf{v}$: $\mathbf{v} = [5; 5]$
$A\mathbf{v} = [3, -1; 2, 4][5; 5] = [3(5) + (-1)(5); 2(5) + 4(5)] = [15 - 5; 10 + 20] = [10; 30]$ Step 3: Calculate the magnitude of the resulting vector $A\mathbf{v}$:
$||A\mathbf{v}|| = \sqrt{10^2 + 30^2} = \sqrt{100 + 900} = \sqrt{1000} = 10\sqrt{10}$ Common Misconceptions: - Choice A ($5\sqrt{2}$) is the magnitude of $\mathbf{v}$ alone, forgetting to apply the transformation matrix A. - Choice D (40) incorrectly sums the components $10 + 30$ instead of computing the Euclidean norm $\sqrt{10^2 + 30^2}$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k? Show your complete algebraic work.

Final Answer: 3. To find k, follow these steps: 1. **Find the vector sum $\mathbf{u} + \mathbf{v}$:** Add corresponding components: $\mathbf{u} + \mathbf{v} = \langle 2 + k, -5 + 3 \rangle = \langle k + 2, -2 \rangle$ 2. **Express the magnitude of $\mathbf{u} + \mathbf{v}$:** The magnitude of a vector $\langle x, y \rangle$ is given by $\sqrt{x^2 + y^2}$: $||\mathbf{u} + \mathbf{v}|| = \sqrt{(k + 2)^2 + (-2)^2} = \sqrt{(k + 2)^2 + 4}$ 3. **Set up and solve the equation:** Given that the magnitude is $\sqrt{29}$: $\sqrt{(k + 2)^2 + 4} = \sqrt{29}$ Square both sides: $(k + 2)^2 + 4 = 29$ $(k + 2)^2 = 25$ Taking the square root of both sides gives $k + 2 = \pm 5$. Since $k > 0$, we take the positive root: $k + 2 = 5 \Rightarrow k = 3$ **Common Traps:** - Choice B (5) results from solving for $k + 2 = 5$ but forgetting to subtract 2 to solve for k. - Choice C (7) results from a sign error in vector addition, writing the horizontal component as $k - 2$, leading to $k - 2 = 5 \Rightarrow k = 7$. - Choice D ($\sqrt{21}$) results from the common algebraic error of expanding $(k + 2)^2$ as $k^2 + 4$, giving $k^2 + 4 + 4 = 29 \Rightarrow k^2 = 21 \Rightarrow k = \sqrt{21}$.

© ClipSAT — clipsat.org | All Rights Reserved

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle -3, 4 \rangle$. What is the vector $2\mathbf{u} + \mathbf{v}$? Show your complete algebraic work.

Final Answer: $\langle 1, -6 \rangle$. To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$: $2\mathbf{u} = 2\langle 2, -5 \rangle = \langle 2(2), 2(-5) \rangle = \langle 4, -10 \rangle$ Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$: $2\mathbf{u} + \mathbf{v} = \langle 4 + (-3), -10 + 4 \rangle = \langle 1, -6 \rangle$ Common misconceptions: - Computing $\mathbf{u} + \mathbf{v}$ without multiplying by 2 yields $\langle 2 + (-3), -5 + 4 \rangle = \langle -1, -1 \rangle$. - Forgetting to multiply the y-component of $\mathbf{u}$ by 2 yields $\langle 4 + (-3), -5 + 4 \rangle = \langle 1, -1 \rangle$. - Subtracting $\mathbf{v}$ instead of adding yields $\langle 4 - (-3), -10 - 4 \rangle = \langle 7, -14 \rangle$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the complex plane, the point representing the complex number $z_1 = 3 + 4i$ is labeled A. The complex number z_2 is obtained by rotating z_1 counterclockwise by 90° about the origin, and its corresponding point is labeled B. The complex number z_3 is the complex conjugate of z_1, and its corresponding point is labeled C. What is the area of ΔABC? Show your complete algebraic work.

Final Answer: 28. 1. **Find point A:** The complex number $z_1 = 3 + 4i$ corresponds to the point $A(3, 4)$ in the complex plane. 2. **Find point B:** Rotating a complex number counterclockwise by 90° about the origin is equivalent to multiplying it by i: $z_2 = i(3 + 4i) = 3i + 4i^2 = -4 + 3i$ Thus, point B has coordinates $(-4, 3)$. 3. **Find point C:** The complex conjugate of $z_1 = 3 + 4i$ is $z_3 = \overline{z_1} = 3 - 4i$. Thus, point C has coordinates $(3, -4)$. 4. **Calculate the area of ΔABC:** Points $A(3, 4)$ and $C(3, -4)$ share the same real coordinate $x = 3$, so segment $\overline{AC}$ is vertical. - Base: $b = 4 - (-4) = 8$ - Height: The horizontal distance from $B(-4, 3)$ to the vertical line $x = 3$ is $h = 3 - (-4) = 7$.
Area $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 7 = 28$ **Common Misconceptions:** - **14:** Arises from using only the half-length of the base (4) rather than the full distance between 4 and -4. - **24:** Arises from misidentifying the coordinates of the rotated point or incorrectly using 6 as the height. - **56:** Arises from forgetting to divide by 2 in the triangle area formula $(8 \times 7 = 56)$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [2, -3; 1, 4]$ and $B = [-5, 1; 3, -2]$, what is the matrix $2A + B$? Show your complete algebraic work.

Final Answer: $[-1, -5; 5, 6]$. To find $2A + B$, first perform scalar multiplication by multiplying each entry of matrix A by 2: $2A = 2[2, -3; 1, 4] = [2(2), 2(-3); 2(1), 2(4)] = [4, -6; 2, 8]$ Next, add matrix B to $2A$ by adding corresponding entries:
$2A + B = [4, -6; 2, 8] + [-5, 1; 3, -2] = [4 + (-5), -6 + 1; 2 + 3, 8 + (-2)] = [-1, -5; 5, 6]$
Common errors: - Simply computing $A + B$ gives $[-3, -2; 4, 2]$. - Computing $2A - B$ gives $[9, -7; -1, 10]$. - Forgetting to multiply one of the entries (such as the bottom-left entry) by 2 yields $[-1, -5; 4, 6]$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the product of $-4i$ and its complex conjugate.

Final Answer: 16. Applying Matrices, Vectors & Complex Numbers principles, $(-4i)(4i) = (0)^2 + (-4)^2 = 16$. Therefore, the verified correct option is 16.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the product of $6 + i$ and its complex conjugate.

Final Answer: 37. Applying Matrices, Vectors & Complex Numbers principles, $(6 + i)(6 - i) = (6)^2 + (1)^2 = 37$. Therefore, the verified correct option is 37.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -20, -21 \rangle$.

Final Answer: 29. $|\vec{v}| = \sqrt{(-20)^2 + (-21)^2} = \sqrt{400 + 441} = 29$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -9, -40 \rangle$.

Final Answer: 41. $|\vec{v}| = \sqrt{(-9)^2 + (-40)^2} = \sqrt{81 + 1600} = 41$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -20, 21 \rangle$.

Final Answer: 29. $|\vec{v}| = \sqrt{(-20)^2 + (21)^2} = \sqrt{400 + 441} = 29$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -24, -7 \rangle$.

Final Answer: 25. $|\vec{v}| = \sqrt{(-24)^2 + (-7)^2} = \sqrt{576 + 49} = 25$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -9, 40 \rangle$.

Final Answer: 41. $|\vec{v}| = \sqrt{(-9)^2 + (40)^2} = \sqrt{81 + 1600} = 41$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -5, 12 \rangle$.

Final Answer: 13. $|\vec{v}| = \sqrt{(-5)^2 + (12)^2} = \sqrt{25 + 144} = 13$.

© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude of the vector $\langle -7, 24 \rangle$.

Final Answer: 25. $|\vec{v}| = \sqrt{(-7)^2 + (24)^2} = \sqrt{49 + 576} = 25$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the magnitude of the vector $\langle -8, 15 \rangle$.

Final Answer: 17. $|\vec{v}| = \sqrt{(-8)^2 + (15)^2} = \sqrt{64 + 225} = 17$.

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

Find the magnitude of the vector $\langle 15, 8 \rangle$.

Final Answer: 17. $|\vec{v}| = \sqrt{(15)^2 + (8)^2} = \sqrt{225 + 64} = 17$.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

Find the magnitude of the vector $\langle 6, 8 \rangle$.

Final Answer: 10. $|\vec{v}| = \sqrt{(6)^2 + (8)^2} = \sqrt{36 + 64} = 10$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

Find the magnitude of the vector $\langle -4, -3 \rangle$.

Final Answer: 5. $|\vec{v}| = \sqrt{(-4)^2 + (-3)^2} = \sqrt{16 + 9} = 5$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

Find the magnitude of the vector $\langle 3, -4 \rangle$.

Final Answer: 5. $|\vec{v}| = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = 5$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

Find the product of $8 + 11i$ and its complex conjugate.

Final Answer: 185. $(8 + 11i)(8 - 11i) = (8)^2 + (11)^2 = 185$.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the product of the complex numbers $(4 - 3i)$ and $(2 + 5i)$, where $i^2 = -1$?

a. $23 + 14i$ ✓

b. $7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

Expand: $4(2) + 4(5i) - 3i(2) - 15i^2 = 8 + 20i - 6i - 15(-1) = 8 + 14i + 15 = 23 + 14i$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the simplified form of the quotient $\frac{5 - i}{2 + i}$?

a. $\frac{9 - 7i}{5}$ ✓

b. $\frac{11 - 7i}{5}$

c. $3 - i$

d. $\frac{9 + 3i}{5}$

Multiply numerator and denominator by conjugate $(2 - i)$: $\frac{(5 - i)(2 - i)}{(2 + i)(2 - i)} = \frac{10 - 5i - 2i + i^2}{4 - i^2} = \frac{10 - 7i - 1}{4 - (-1)} = \frac{9 - 7i}{5}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of i^{47}, where $i = \sqrt{-1}$?

a. 1

b. -1

c. i

d. $-i$ ✓

Divide power by 4: $47 = 4(11) + 3$. Thus $i^{47} = (i^4)^{11} \cdot i^3 = 1 \cdot (-i) = -i$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the modulus (absolute value) of the complex number $z = -5 + 12i$?

a. 7

b. 13 ✓

c. 17

d. $\sqrt{119}$

Modulus $|z| = \sqrt{a^2 + b^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the distance in the complex plane between $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$?

a. $2\sqrt{13}$ ✓

b. $\sqrt{20}$

c. 10

d. 5

Distance $= |z_1 - z_2| = |(3 - (-1)) + (-2 - 4)i| = |4 - 6i| = \sqrt{4^2 + (-6)^2} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z = 2 + 2i$, what is the argument θ of z in radians, where $0 \le \theta < 2\pi$?

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$ ✓

c. $\frac{\pi}{3}$

d. $\frac{3\pi}{4}$

Both real and imaginary parts are positive (Quadrant I). $\tan\theta = \frac{b}{a} = \frac{2}{2} = 1 \Rightarrow \theta = \frac{\pi}{4}$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given matrices $A = [[3, -1], [2, 4]]$ and $B = [[1, 5], [-2, 3]]$, what is the matrix $2A - B$?

a. $[[5, -7], [6, 5]]$ ✓

b. $[[5, 3], [2, 5]]$

c. $[[7, -7], [6, 11]]$

d. $[[5, -7], [2, 5]]$

$2A = [[6, -2], [4, 8]]$. Subtract B: $[[6 - 1, -2 - 5], [4 - (-2), 8 - 3]] = [[5, -7], [6, 5]]$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the determinant of the 2×2 matrix $M = [[4, 6], [-2, 3]]$?

a. 0

b. 12

c. 24 ✓

d. 36

Determinant $\det(M) = ad - bc = 4(3) - (6)(-2) = 12 - (-12) = 24$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k will the matrix $K = [[k, 6], [2, 3]]$ NOT have an inverse?

a. 1

b. 2

c. 3

d. 4 ✓

A matrix is singular (no inverse) when its determinant is zero:

$k(3) - (6)(2) = 0 \Rightarrow 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$ are given. What are the components of $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$ ✓

b. $\langle 1, 1 \rangle$

c. $\langle 4, 3 \rangle$

d. $\langle -1, -3 \rangle$

$2\mathbf{u} = \langle 6, -8 \rangle$. Adding $\mathbf{v}$: $\langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the dot product of the vectors $\mathbf{a} = \langle 5, -2 \rangle$ and $\mathbf{b} = \langle 4, 7 \rangle$?

a. 6 ✓

b. 26

c. -6

d. 34

$\mathbf{a} \cdot \mathbf{b} = (5)(4) + (-2)(7) = 20 - 14 = 6$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the magnitude of the vector $\mathbf{w} = \langle -9, 12 \rangle$?

a. 13

b. 15 ✓

c. 17

d. 21

Magnitude $||\mathbf{w}|| = \sqrt{(-9)^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $-4 + 5i$ and its complex conjugate.

a. 52

b. 8

c. **41** ✓

d. 35

$(-4 + 5i)(-4 - 5i) = (-4)^2 + (5)^2 = 41.$

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $3 - 2i$ and its complex conjugate.

a. **13** ✓

b. 34

c. 1

d. 27

$(3 - 2i)(3 + 2i) = (3)^2 + (-2)^2 = 13.$

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $4 - 3i$ and its complex conjugate.

a. 53

b. 11

c. 42

d. **25** ✓

$(4 - 3i)(4 + 3i) = (4)^2 + (-3)^2 = 25.$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the product of $7 + 7i$ and its complex conjugate.

a. 111

b. **98** ✓

c. 123

d. 71

$(7 + 7i)(7 - 7i) = (7)^2 + (7)^2 = 98.$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the complex number system, for what real value of k is the expression $\frac{k+8i}{2-5i}$ a purely imaginary number? (Note: $i = \sqrt{-1}$)

a. -20

b. $-\frac{16}{5}$

c. $\frac{16}{5}$

d. 20 ✓

To determine when the fraction is purely imaginary, multiply the numerator and denominator by the complex conjugate of the denominator, $2+5i$: $\frac{k+8i}{2-5i} \cdot \frac{2+5i}{2+5i} = \frac{(k+8i)(2+5i)}{(2)^2-(5i)^2}$ Simplify the denominator: $2^2 - 25i^2 = 4 - 25(-1) = 4 + 25 = 29$ Expand the numerator using FOIL: $(k+8i)(2+5i) = 2k + 5ki + 16i + 40i^2 = 2k + (5k+16)i - 40 = (2k-40) + (5k+16)i$ Thus, the expression can be written in standard form $a+bi$ as: $\frac{2k-40}{29} + \left(\frac{5k+16}{29}\right)i$ A complex number is purely imaginary if its real part equals 0 and its imaginary part is nonzero: $\frac{2k-40}{29} = 0 \Rightarrow 2k - 40 = 0 \Rightarrow 2k = 40 \Rightarrow k = 20$ When $k = 20$, the imaginary part is $\frac{5(20)+16}{29} = \frac{116}{29} = 4eq0$, yielding the purely imaginary number $4i$. Common traps: - Setting the imaginary part to 0 yields $5k + 16 = 0 \Rightarrow k = -\frac{16}{5}$, which makes the expression a purely real number rather than purely imaginary. - Forgetting that $i^2 = -1$ leads to $2k + 40 = 0 \Rightarrow k = -20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

In the complex number system, where $i^2 = -1$, the expression $\frac{a+6i}{2-i}$ is equal to $4 + bi$ for some real numbers a and b. What is the value of $a + b$?

a. 5

b. 8

c. 13

d. **18** ✓

To find the value of $a + b$, first relate the two sides of the equation: $\frac{a+6i}{2-i} = 4 + bi$ Multiply both sides by $2 - i$: $a + 6i = (4 + bi)(2 - i)$ Expand the right-hand side using the FOIL method and the identity $i^2 = -1$: $(4 + bi)(2 - i) = 8 - 4i + 2bi - bi^2$ Because $i^2 = -1$, the term $-bi^2 = -b(-1) = +b$: $8 - 4i + 2bi + b = (8 + b) + (2b - 4)i$ Now equate the corresponding real and imaginary components from both sides of $a + 6i = (8 + b) + (2b - 4)i$: 1. Equating imaginary parts: $6 = 2b - 4 \Rightarrow 2b = 10 \Rightarrow b = 5$ 2. Equating real parts: $a = 8 + b \Rightarrow a = 8 + 5 = 13$ Finally, determine $a + b$: $a + b = 13 + 5 = 18$ Common traps: - Choice A (5): Solving only for b and stopping early. - Choice B (8): Making a sign error with i^2 by treating $-bi^2$ as $-b$, resulting in $a = 8 - 5 = 3$ and $a + b = 8$. - Choice C (13): Solving only for a and stopping early.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equal to $(5 - 3i) - (2 - 8i)$?

a. $3 + 5i$ ✓

b. $3 - 11i$

c. $7 - 11i$

d. $3 - 5i$

To subtract complex numbers, distribute the negative sign to both terms inside the second set of parentheses: $(5 - 3i) - (2 - 8i) = 5 - 3i - 2 + 8i$ Next, combine the real parts and the imaginary parts separately: Real parts: $5 - 2 = 3$ Imaginary parts: $(-3 + 8)i = 5i$ Combining these yields $3 + 5i$. Common Misconceptions: - $3 - 11i$ occurs if the negative sign is not distributed to $-8i$, yielding $-3i - 8i = -11i$. - $7 - 11i$ occurs if the real parts are added ($5 + 2 = 7$) instead of subtracted. - $3 - 5i$ occurs from an arithmetic sign error when combining $-3 + 8$.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

If $A = (2, -11; 9, -3)$ and $B = (-14, -11; 11, -5)$, find the entry in row 1, column 2 of $A + B$.

a. -25

b. -21

c. -16

d. **-22** ✓

Applying Matrices, Vectors & Complex Numbers principles, Row 1, Col 2: $-11 + -11 = -22$. Therefore, the verified correct option is -22.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

If $M = (-3, -5; 6, -3)$, find the entry in row 2, column 1 of $2M$.

a. 16

b. 14

c. **12** ✓

d. 11

Applying Matrices, Vectors & Complex Numbers principles, Row 2, Col 1: $2 \times (6) = 12$. Therefore, the verified correct option is 12.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

If $M = (-5, -1; 2, -4)$, find the entry in row 2, column 1 of $5M$.

a. 16

b. 15

c. **10** ✓

d. 8

Applying Matrices, Vectors & Complex Numbers principles, Row 2, Col 1: $5 \times (2) = 10$. Therefore, the verified correct option is 10.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

If $M = (-5, -5; 4, -8)$, find the entry in row 2, column 1 of $4M$.

a. **16** ✓

b. 10

c. 18

d. 13

Applying Matrices, Vectors & Complex Numbers principles, Row 2, Col 1: $4 \times (4) = 16$. Therefore, the verified correct option is 16.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Functions & Graphs

If $M = (-6, 2; 3, -7)$, find the entry in row 2, column 1 of $5M$.

a. 13

b. 10

c. 14

d. **15** ✓

Applying Matrices, Vectors & Complex Numbers principles, Row 2, Col 1: $5 \times (3) = 15$. Therefore, the verified correct option is 15.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

If $M = (3, -7; 5, -1)$, find the entry in row 2, column 1 of $3M$.

a. **15** ✓

b. 8

c. 14

d. 17

Applying Matrices, Vectors & Complex Numbers principles, Row 2, Col 1: $3 \times (5) = 15$. Therefore, the verified correct option is 15.

© ClipSAT — clipsat.org | All Rights Reserved